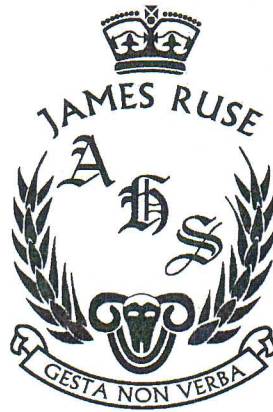


Name:	
Class:	



YEAR 12

ASSESSMENT TEST 2

TERM 1, 2016

MATHEMATICS

EXTENSION 2

*Time Allowed – 120 Minutes
(Plus 5 minutes Reading Time)*

General Instructions:

- All questions may be attempted
- All questions are of equal value
- Reference Sheet will be supplied
- Department of Education approved calculators and templates are permitted
- In every Question, show all relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for careless or badly arranged work

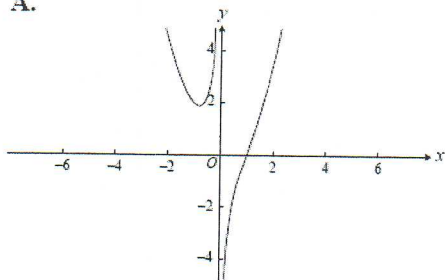
**Question 1-5 are to be completed on multiple choice sheet then Question 6,7,8,9 to be completed on separate sheets of paper and handed in in separate bundles.
Each question must show your Candidate Number (in the top right hand corner).**

For the following questions colour the most correct answer on your multiple choice answer sheet.

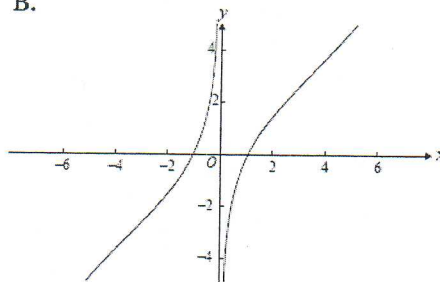
1. Let $f(x) = \frac{x^k + a}{x}$, where k and a are real constants.

If k is an odd integer which is greater than 1 and $a < 0$, a possible graph of $f(x)$ could be

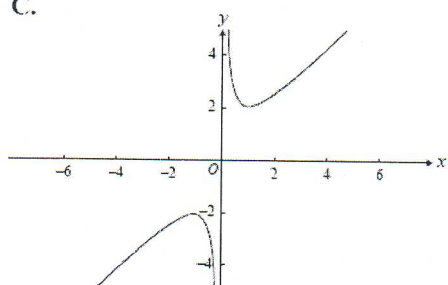
A.



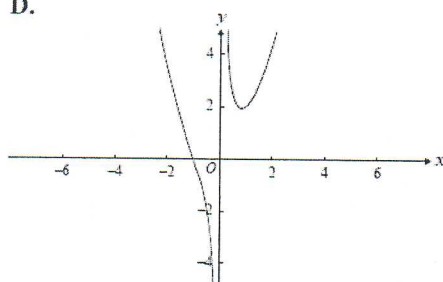
B.



C.



D.



2. The coordinates of the foci of the hyperbola $\frac{x^2}{4} - y^2 = 1$ are given by:

A. $(0, \pm\sqrt{5})$

B. $(\pm\sqrt{5}, 0)$

C. $(0, \pm 2\sqrt{5})$

D. $(\pm 2\sqrt{5}, 0)$

3. The polynomial $P(z) = z^3 + (1+i)z^2 + (1+i)z + 1$ has a real zero $z = -1$ and a complex zero $z = \alpha$. By considering the sum and the product of the roots, the third root is:

A. $\frac{1}{\alpha}$

B. $\bar{\alpha}$

C. $-\alpha$

D. $(1-\alpha)$

4. $\int x \sin 2x \, dx =$

A. $-2x \cos 2x + \sin 2x + C$

C. $\frac{x}{2} \cos 2x + \frac{1}{4} \sin 2x + C$

B. $-\frac{x}{2} \cos 2x - \frac{1}{4} \sin 2x + C$

D. $-\frac{x}{2} \cos 2x + \frac{1}{4} \sin 2x + C$

5. The graph of $y = \frac{1}{ax^2 + bx + c}$ has asymptotes at $x = -5$, $x = 3$ and $y = 0$.

Given that the graph has one stationary point with a y -coordinate of $-\frac{1}{8}$, it follows that

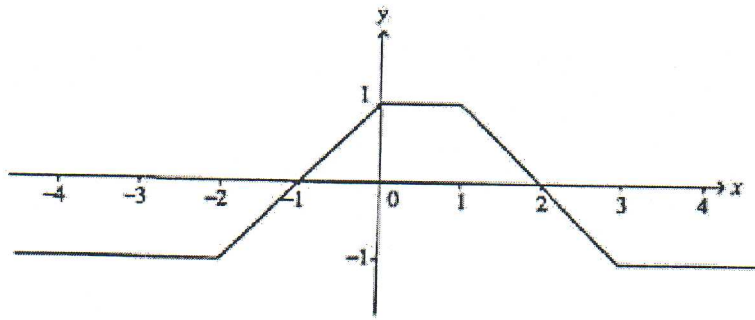
A. $a = 1$, $b = 2$, $c = -15$

B. $a = \frac{1}{2}$, $b = 1$, $c = -\frac{15}{2}$

C. $a = -\frac{1}{2}$, $b = -1$, $c = 15$

D. $a = \frac{1}{2}$, $b = -1$, $c = -\frac{15}{2}$

- a) The diagram is a sketch of the function $y = f(x)$.



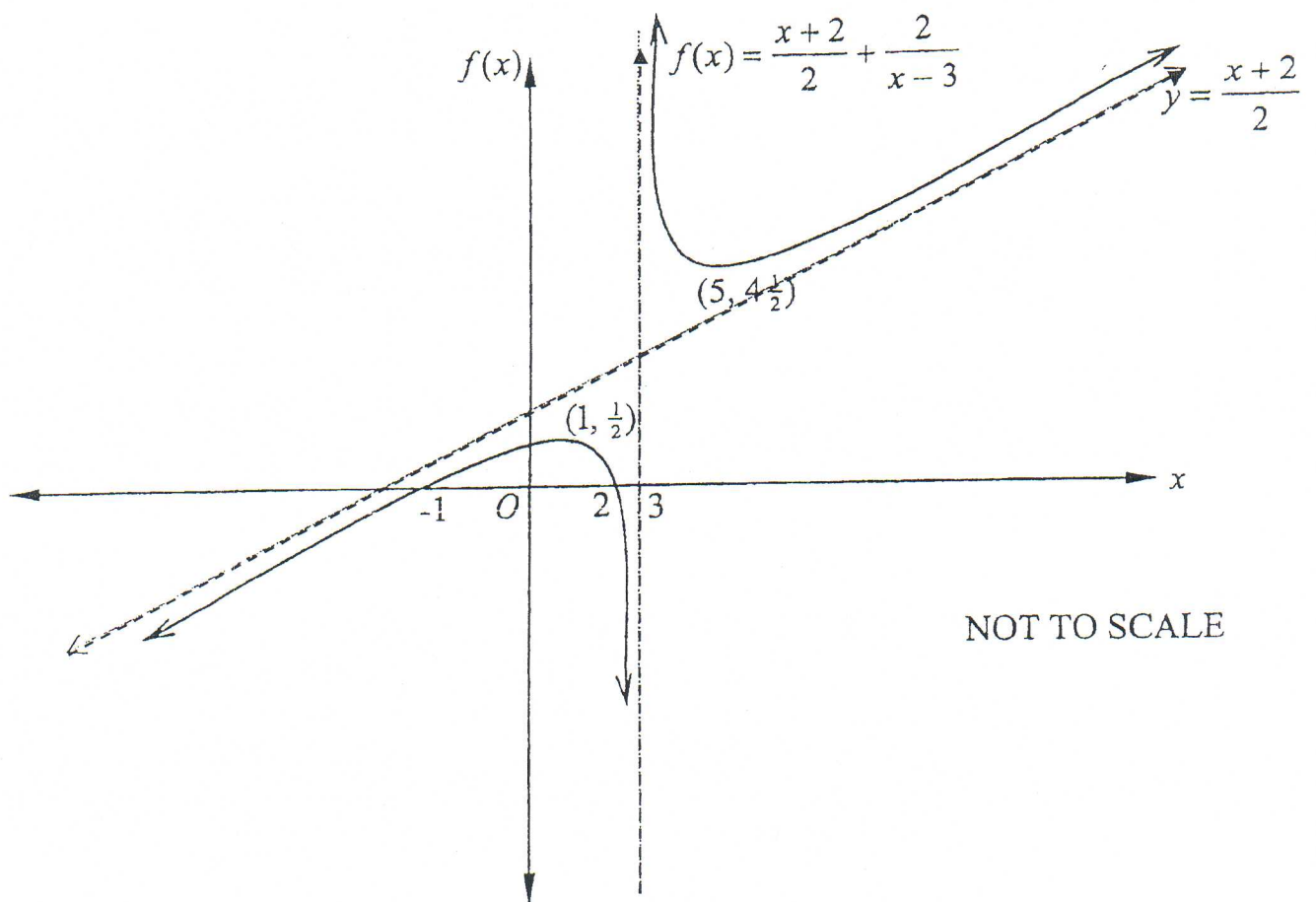
On separate diagrams sketch:

- | | | |
|-------|-----------------------|---|
| (i) | $y = \frac{1}{f(x)}$ | 2 |
| (ii) | $y = f(x) - f(x) $ | 2 |
| (iii) | $y = \sin^{-1}(f(x))$ | 2 |
| (iv) | $y = e^{f(x)}$ | 2 |

b) Consider the function $f(x) = \begin{cases} \frac{e^x - 1}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$

- | | | |
|-------|--|---|
| (i) | Use differentiation to show that $e^{-x} + x - 1 \geq 0$ for all values of x . Hence show that $f(x)$ is an increasing function for $x \neq 0$. | 3 |
| (ii) | Show that $f(x)$ is continuous at $x = 0$. | 2 |
| (iii) | Sketch the graph of $y = f(x)$. | 1 |

- c) The function defined by $f(x) = \frac{x+2}{2} + \frac{2}{x-3}$ is drawn below.



Draw a separate sketch of $y = [f(x)]^2$.

2

a) Find $\int \frac{\sin 2x}{(1 + \sin^2 x)^2} dx$ 1

b) Consider $I = \int_{-1}^1 \frac{x^2 e^x}{e^x + 1} dx$ and $J = \int_{-1}^1 \frac{x^2}{e^x + 1} dx$.

(i) Use the substitution $u = -x$ in I to show that $I = J$. 3

(ii) Hence evaluate I . 1

c) Evaluate $\int_0^{\frac{\pi}{4}} \frac{1 - \tan x}{1 + \tan x} dx$ 3

d) (i) Use the result $\int_0^a f(x) dx = \int_0^a f(a - x) dx$, to show that

$$\int_0^{\frac{\pi}{4}} \ln(1 + \tan x) dx = \int_0^{\frac{\pi}{4}} \ln\left(\frac{2}{1 + \tan x}\right) dx. \quad 2$$

(i) Hence, evaluate $\int_0^{\frac{\pi}{4}} \ln(1 + \tan x) dx$. 2

e) Using a suitable substitution find the exact value of $\int_0^1 \sqrt{2 - \sqrt{x}} dx$. 4

a) The locus of a point is defined by the equation $|z - 2| = 2 \operatorname{Re}\left(z - \frac{1}{2}\right)$.

- (i) If $z = x + iy$, explain why $x \geq \frac{1}{2}$. 1
- (ii) Show that the locus satisfies the equation $3x^2 - y^2 = 3$. 2
- (iii) Describe the locus that satisfies (i) and (ii). 1
- (iv) Sketch the locus showing its asymptotes and vertex. 1
- (v) Find all possible values for $\arg z$ for this locus. 1

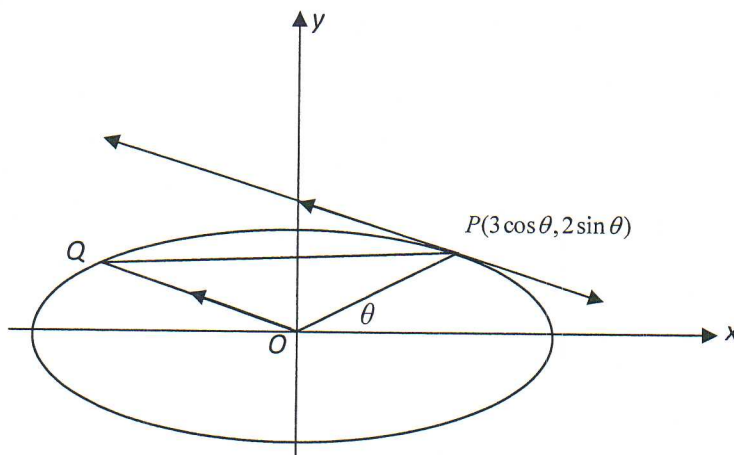
b) The equation $3x^3 - 9x^2 + 6x + 2 = 0$ has roots α , β and γ .

- (i) Find the cubic equation with roots α^2 , β^2 and γ^2 . 2
- (ii) Hence evaluate $\alpha^3\beta\gamma + \alpha\beta^3\gamma + \alpha\beta\gamma^3$. 2

c) Given that $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$,

- (i) By letting $x = \frac{2}{3}\cos \theta$, use the above formula to find the three roots of the equation $27x^3 - 9x + 1 = 0$. 3
- (ii) Hence, find the value of $\cos \frac{\pi}{9} \cos \frac{3\pi}{9} \cos \frac{5\pi}{9} \cos \frac{7\pi}{9}$. 3

The point $P(3 \cos \theta, 2 \sin \theta)$ lies on the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ with centre O .



- (i) Find the equation of the tangent to this ellipse at P . 2
- (ii) A line passing through the origin O parallel to the tangent at P meets the ellipse at Q as shown on the diagram. Show that Q is $(-3 \sin \theta, 2 \cos \theta)$. 2
- (iii) Show that the distance between the tangent at P and the line OQ is $\frac{6}{OQ}$. 3
- (iv) Hence prove that the area of $\triangle OPQ$ is independent of the position of P . 1
- (v) Find the length of PQ and state the maximum and minimum values of this length for $-\pi \leq \theta \leq \pi$. 3
- (vi) The tangents at P and Q to the ellipse intersect at T . Find the coordinates of T , and verify that T lies on the curve $4x^2 + 9y^2 = 72$. 2
- (v) PD is a diameter of the ellipse. The tangent at P meets the vertical through D at V . Show that the ratio of the area of $\triangle PDV$ to the area of the ellipse is $2 : \pi |\tan \theta|$. 3

Questions continue over page...

- a) i) Show that, if $(r \cos \theta, r \sin \theta)$ and $\left(s \cos\left(\theta + \frac{\pi}{2}\right), s \sin\left(\theta + \frac{\pi}{2}\right)\right)$ lie on the 2

hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with centre O , then $\frac{1}{r^2} + \frac{1}{s^2} = \frac{1}{a^2} - \frac{1}{b^2}$.

- (ii) Deduce that, if P and Q are points on the hyperbola such that OP is 3

perpendicular to OQ , then $\frac{1}{OP^2} + \frac{1}{OQ^2}$ is independent of the position of

P and Q . Give a sketch of the hyperbola and points O, P, Q related in this way.

- b) Find the point(s) on the curve $xy(x+y) + 16 = 0$ where the gradient of the tangent 4
to the curve is -1.

c) $I_n = \int_0^1 x^n \ln(1+x) dx, n = 0, 1, 2, \dots$

- (i) Show that $\int \ln(1+x) dx = (1+x) \ln(1+x) - x + c$. 1

- (ii) Show that $(n+1)I_n = 2 \ln 2 - \frac{1}{n+1} - nI_{n-1}, n = 1, 2, \dots$ 2

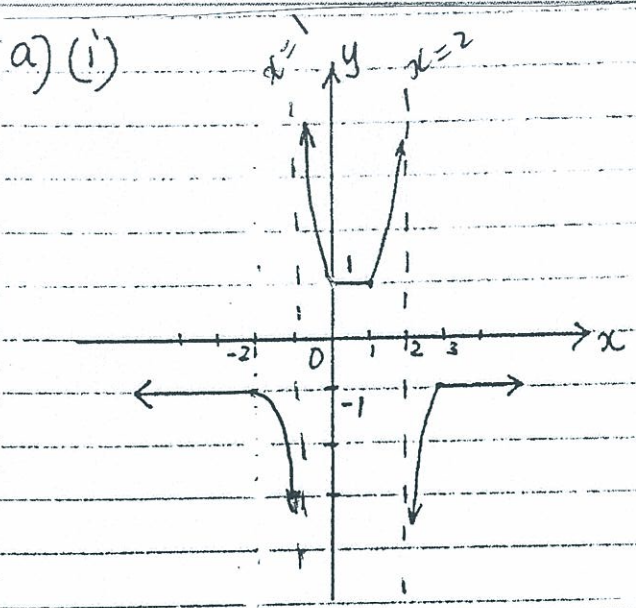
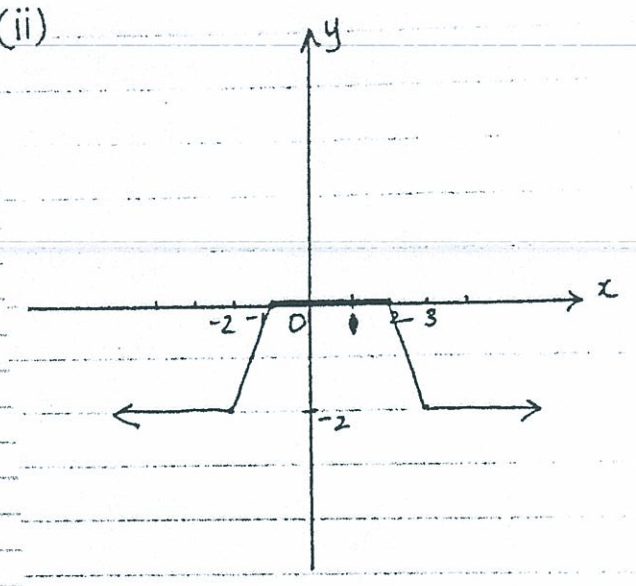
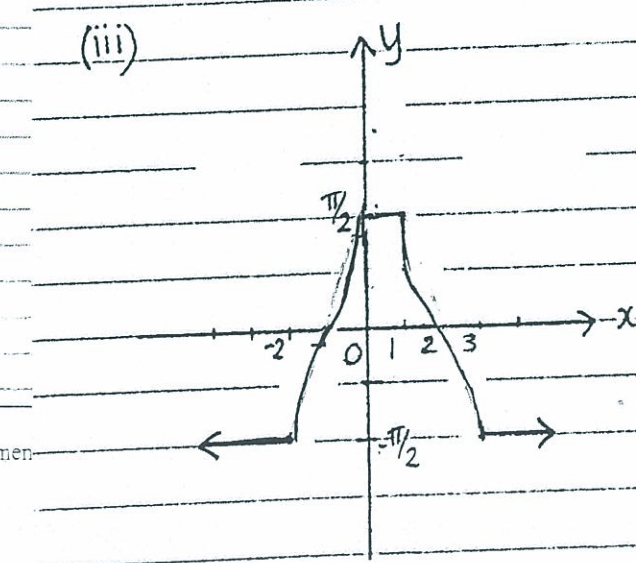
- (iii) Evaluate $3I_2$ and $4I_3$. 2

- (iv) Show that $(n+1)I_n = \begin{cases} \frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{n+1}, & n \text{ odd} \\ 2 \ln 2 - \left(\frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{n+1} \right), & n \text{ even} \end{cases}$ 2

Ext 2 MATHEMATICS: Question MLC

Suggested Solutions	Marks	Marker's Comments
1. crosses x-axis when $x^k + a = 0$ $x^k = -a$ but a is negative so has to be graph A	A	
2. $a = 2$ $b = 1$ $1 = 4(e^2 - 1)$ $\frac{1}{4} = e^2 - 1$ $e^2 = \frac{5}{4}$ $e = \pm \sqrt{\frac{5}{4}}$ $e = \pm \frac{\sqrt{5}}{2}$	B	
$S(\pm\sqrt{5}, 0)$		
3. $\alpha\beta\gamma = -1$ $-1 \times \alpha\beta = -1$ $\alpha\beta = 1 \Rightarrow \beta = \frac{1}{\alpha}$	A	
4. $\int x \sin 2x dx = \int x \frac{d}{dx} \left(-\frac{1}{2} \cos 2x \right) dx$ $= -\frac{1}{2} x \cos 2x + \frac{1}{2} \int \cos 2x dx$ $= -\frac{1}{2} x \cos 2x + \frac{1}{4} \sin 2x + c$	D	
5. $y = \frac{1}{d(x+5)(1+3)}$ $\frac{dy}{dx} = \frac{-(2x+2)c}{d(x^2+2x-15)^2}$ start point when $x = -1$ $y = -\frac{1}{8}$ when $x = -1 \Rightarrow y = \frac{1}{d(-16)}$ $\Rightarrow d = \frac{1}{2}$ ($a = \frac{1}{2}$ $b = 1$ $c = -\frac{\sqrt{5}}{2}$)	B	

MATHEMATICS: Question..... Ex 2 Q 6

Suggested Solutions	Marks	Marker's Comments
a) (i) 	②	① Asymptotes and curved lines ① straight lines
(ii) 	②	① graph $-1 \leq x \leq 2$ ① complete graph
(iii) 	②	① straight sections ① curved sections (shape of inverse sine function)

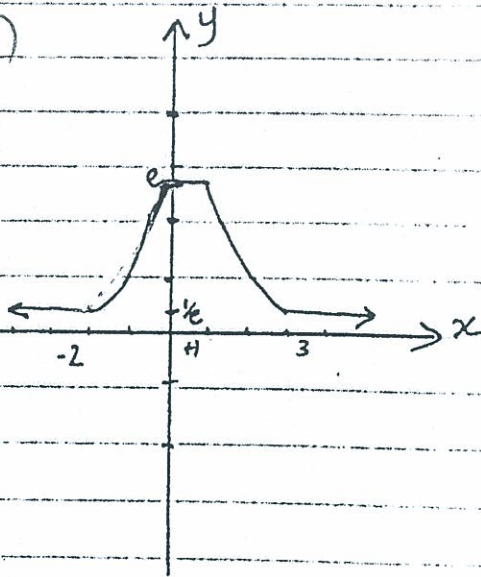
MATHEMATICS: Question... EXT2 Q6

Suggested Solutions

Marks

Marker's Comments

a) (iv)



②

① straight sections
① curved sections
(shape e^x and e^{-x})

$$b) f(x) = \begin{cases} \frac{e^x - 1}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

(i) Consider the function $g(x) = e^{-x} + x - 1$.
 $g(0) = 0$ and $g'(x) = -e^{-x} + 1$
 $\therefore g'(0) = 0$

Also, $g''(x) = e^{-x} > 0$ for all x . and $g(0) = 0$

$\therefore g(x)$ has a minimum of 0 when $x = 0$.

$\therefore e^{-x} + x - 1 \geq 0$ for all x , with equality only if $x = 0$.

$$\text{For } x \neq 0, f'(x) = \frac{d}{dx} \left(\frac{e^x - 1}{x} \right)$$

$$= \frac{e^x \cdot x - (e^x - 1) \cdot 1}{x^2}$$

$$= \frac{1 + xe^x - e^x}{x^2}$$

$$= \frac{e^x}{x^2} (e^{-x} + x - 1)$$

$$> 0 \text{ as } e^x > 0, x^2 > 0, e^{-x} + x - 1 > 0$$

Hence $f(x)$ is an increasing function.

③

① full + complete proof of $e^{-x} + x - 1 \geq 0$ by showing minimum at (0,0)

$$① f'(x) = 1 + \frac{xe^x - e^x}{x^2}$$

① explanation that $f'(x) > 0$

MATHEMATICS: Question EXT 2 Q6

Suggested Solutions

Marks

Marker's Comments

(ii) Let $h(x) = e^x$. Then $h'(x) = e^x$.

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \lim_{x \rightarrow 0} \frac{e^x - e^0}{x - 0}$$

$$= \frac{h'(0)}{1}$$

$$= 1$$

$$= f'(0)$$

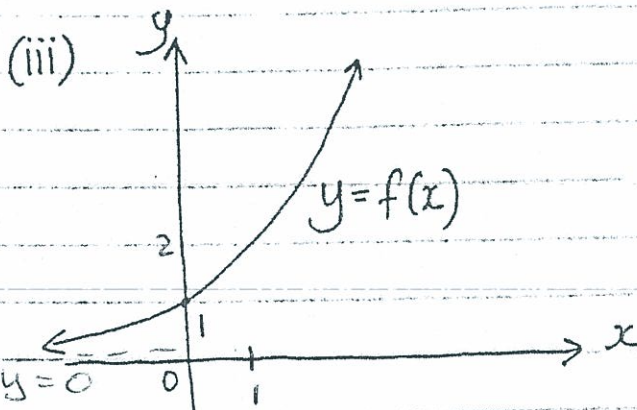
$\therefore f(x)$ is continuous at $x=0$.

(2)

① Apply concept
 $\lim_{x \rightarrow 0} f(x) = f(0) = 1$
 $x \rightarrow 0$

① complete proof

$$\lim_{x \rightarrow 0} \left[\frac{e^x - 1}{x} \right] = 1$$

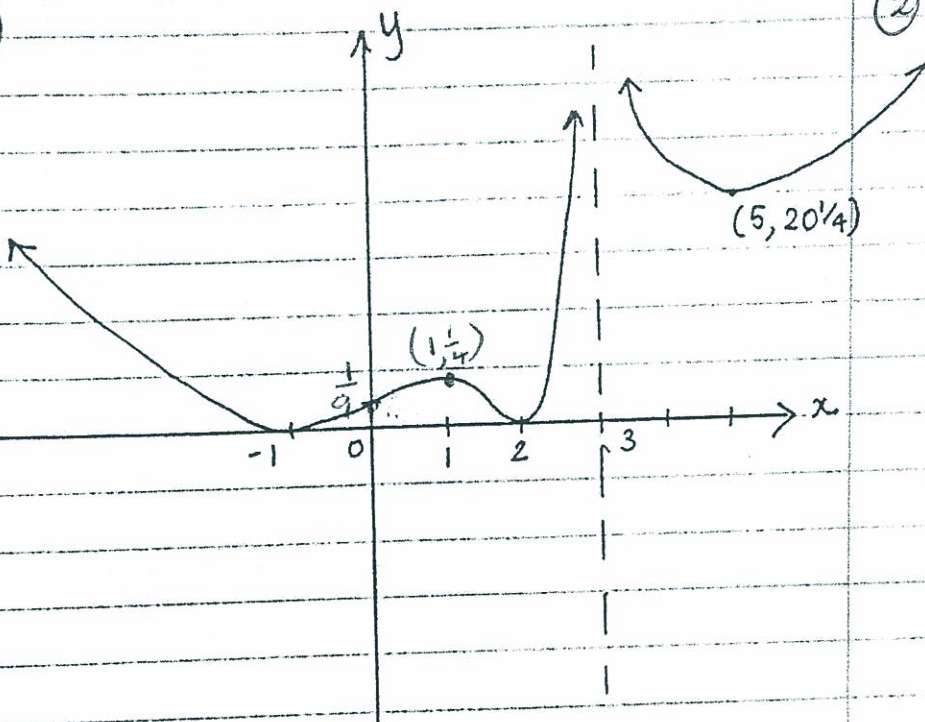


(1)

Must show
increasing
graph with
increasing
gradient
through $y=1$

Asymptote $y=0$
x scale should
be labelled

6c)



(2)

① Rounded.
T.P.'s at
 $x=-1$ and $x=2$.

① Rest of graph.
T.P.'s must
be labelled
with coordinates
y intercept
should be shown

Question 7

a) $\int \frac{\sin 2x}{(1 + \sin^2 x)^2} dx$

Let $u = \sin^2 x$

$$\frac{du}{dx} = 2 \sin x \cos x$$

$$= \sin 2x$$

$$\therefore \int \frac{\sin 2x}{(1 + \sin^2 x)^2} dx = \int \frac{du}{(1 + u)^2}$$

$$= -\frac{1}{1 + u} + C$$

$$= -\frac{1}{1 + \sin^2 x} + C$$

Note: Other answers include:

$$\ast \frac{-2}{3 - \cos 2x} + C$$

$$\ast \frac{-1}{2 - \cos^2 x} + C$$

b) (i) Let $u = -x$

$$\frac{du}{dx} = -1$$

$$I = \int_{-1}^1 \frac{x^2 e^x}{e^x + 1} dx$$

$$= \int_{+1}^{-1} \frac{u^2 e^{-u}}{e^{-u} + 1} du$$

(1)

$$= \int_{-1}^1 \frac{u^2}{e^u} \cdot \frac{e^u}{1 + e^u} du$$

$$= \int_{-1}^1 \frac{u^2}{1 + e^u} du$$

(1)

$$= \int_{-1}^1 \frac{x^2}{1 + e^x} dx$$

(1)

$$= J$$

* Students should state (change of variable) in the last step

$$(ii) \quad I + J = \int_{-1}^1 \left[\frac{x^2 e^x}{e^x + 1} + \frac{x^2}{e^x + 1} \right] dx$$

$$2I = \int_{-1}^1 x^2 \frac{(e^x + 1)}{e^x + 1} dx$$

$$= \int_{-1}^1 x^2 dx$$

$$= \left[\frac{1}{3} x^3 \right]_{-1}^1$$

$$= \frac{1}{3} (1 - (-1)) = \frac{2}{3} \quad \text{--- (1)}$$

$$\therefore I = \frac{1}{3}$$

$$\text{Since } \int_0^a f(a-x) dx = \int_0^a f(x) dx$$

$$c) \quad \int_0^{\pi/4} \frac{1 - \tan x}{1 + \tan x} dx \quad \xrightarrow{\text{or}} \quad \int_0^{\pi/4} \frac{1 - \tan(\frac{\pi}{4} - x)}{1 + \tan(\frac{\pi}{4} - x)} dx \quad \text{--- (1)}$$

$$= \int_0^{\pi/4} \frac{1 - \frac{\sin x}{\cos x}}{1 + \frac{\sin x}{\cos x}} dx$$

$$= \int_0^{\pi/4} \frac{1 - \frac{1 - \tan x}{1 + \tan x}}{1 + \frac{1 - \tan x}{1 + \tan x}} dx$$

$$= \int_0^{\pi/4} \frac{\cos x - \sin x}{\cos x + \sin x} dx$$

$$= \int_0^{\pi/4} \frac{1 + \tan x - 1 + \tan x}{1 + \tan x + 1 - \tan x} dx$$

$$= \left[\ln(\cos x + \sin x) \right]_0^{\pi/4}$$

$$= \int_0^{\pi/4} \tan x dx$$

$$= \ln\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right) - \ln(1+0)$$

$$= \left[-\ln |\cos x| \right]_0^{\pi/4}$$

$$= \ln \sqrt{2} \quad \text{--- (1)}$$

$$= \frac{1}{2} \ln 2 \quad \text{or} \quad \ln \sqrt{2}$$

$$= -\ln \frac{1}{\sqrt{2}} + \ln 1$$

$$= \ln \sqrt{2} \quad \text{--- (1)}$$

$$d)(i) \int_0^{\pi/4} \ln(1+\tan x) dx$$

$$= \int_0^{\pi/4} \ln(1+\tan(\frac{\pi}{4}-x)) dx \quad \text{since } \int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$= \int_0^{\pi/4} \ln\left(1 + \frac{\tan \frac{\pi}{4} - \tan x}{1 + \tan \frac{\pi}{4} \tan x}\right) dx \quad \text{--- (1)}$$

$$= \int_0^{\pi/4} \ln\left(1 + \frac{1 - \tan x}{1 + \tan x}\right) dx$$

$$= \int_0^{\pi/4} \ln\left(\frac{1 + \tan x + 1 - \tan x}{1 + \tan x}\right) dx$$

$$= \int_0^{\pi/4} \ln\left(\frac{2}{1 + \tan x}\right) dx \quad \text{--- (2)}$$

$$(ii) \int_0^{\pi/4} \ln(1+\tan x) dx$$

$$= \int_0^{\pi/4} \ln\left(\frac{2}{1+\tan x}\right) dx$$

$$= \int_0^{\pi/4} \ln 2 dx - \int_0^{\pi/4} \ln(1+\tan x) dx \quad \text{--- (3)}$$

$$\therefore 2 \int_0^{\pi/4} \ln(1+\tan x) dx = \int_0^{\pi/4} \ln 2 dx$$

$$= [x \ln 2]_0^{\pi/4}$$

$$= \frac{\pi}{4} \ln 2 - 0$$

$$\therefore \int_0^{\pi/4} \ln(1+\tan x) dx = \frac{\pi}{8} \ln 2 \quad \text{--- (4)}$$

$$e) \int_0^1 \sqrt{2-\sqrt{x}} \, dx$$

$$\text{let } u = 2 - \sqrt{x}$$

$$\text{when } x=1, \quad u=1$$

$$\therefore \sqrt{x} = 2 - u$$

$$x=0, \quad u=2$$

$$x = 4 - 4u + u^2$$

$$\frac{dx}{du} = -4 + 2u$$

$$\therefore I = \int_2^1 \sqrt{u} (-4 + 2u) du \quad \text{--- (i)}$$

$$= \int_2^1 2u^{\frac{3}{2}} - 4u^{\frac{1}{2}} du$$

$$\text{--- (1)}$$

$$= 2 \left[\frac{u^{\frac{5}{2}}}{\frac{5}{2}} \right]_2^1 - 4 \left[\frac{u^{\frac{3}{2}}}{\frac{3}{2}} \right]_2^1 \quad \text{--- (i)}$$

$$= 2 \left(\frac{2}{5} - \frac{8\sqrt{2}}{5} \right) - 4 \left(\frac{2}{3} - \frac{4\sqrt{2}}{3} \right)$$

$$= \left(\frac{4}{5} - \frac{8}{3} \right) + \left(\frac{16\sqrt{2}}{3} - \frac{16\sqrt{2}}{5} \right)$$

$$= \frac{32\sqrt{2}}{15} - \frac{28}{15} \quad \text{--- (1)}$$

$$e) \int_0^1 \sqrt{2-\sqrt{x}} dx$$

$$\text{Let } x = 4 \sin^4 \theta$$

$$\frac{dx}{d\theta} = 16 \sin^3 \theta \cos \theta$$

$$\text{At } x=0, \theta=0$$

$$x=1, \theta=\pi/4$$

①

$$\int_0^1 \sqrt{2-\sqrt{x}} dx = \int_0^{\pi/4} \sqrt{2-2\sin^2 \theta} \cdot 16 \sin^3 \theta \cos \theta d\theta$$

$$= \int_0^{\pi/4} \sqrt{2} \sqrt{1-\sin^2 \theta} \cdot 16 \sin^3 \theta \cos \theta d\theta$$

$$= \sqrt{2} \int_0^{\pi/4} \cos \theta \cdot 16 \sin^3 \theta \cos \theta d\theta$$

$$= 16\sqrt{2} \int_0^{\pi/4} \sin^3 \theta \cos^2 \theta d\theta \quad \text{--- ①}$$

$$= 16\sqrt{2} \int_0^{\pi/4} \sin \theta \cos^2 \theta (1-\cos^2 \theta) d\theta$$

$$= 16\sqrt{2} \int_0^{\pi/4} \cos^2 \theta \sin \theta d\theta$$

$$- 16\sqrt{2} \int_0^{\pi/4} \cos^4 \theta \sin \theta d\theta$$

$$= 16\sqrt{2} \left[\frac{1}{3} \cos^3 \theta + \frac{1}{5} \cos^5 \theta \right]_0^{\pi/4} \quad \text{--- ①}$$

$$= 16\sqrt{2} \left(-\frac{1}{3} \left(\frac{1}{\sqrt{2}} \right)^3 + \frac{1}{5} \left(\frac{1}{\sqrt{2}} \right)^5 \right) + 16\sqrt{2} \left(\frac{1}{3} - \frac{1}{5} \right)$$

$$= 16\sqrt{2} \left(-\frac{1}{3} \times \frac{1}{2\sqrt{2}} + \frac{1}{5} \times \frac{1}{4\sqrt{2}} \right) + 16\sqrt{2} \left(\frac{2}{15} \right)$$

$$= \left(-\frac{16}{6} + \frac{16}{20} \right) + \frac{32\sqrt{2}}{15}$$

$$= -\frac{28}{15} + \frac{32\sqrt{2}}{15} = \frac{-28+32\sqrt{2}}{15} \quad \text{--- ①}$$

$$Q1) \int_0^1 \sqrt{2-\sqrt{x}} \, dx$$

$$= \int_0^1 \sqrt{2-u} \, 2u \, du \quad \text{--- (1)}$$

$$= 2 \int_0^1 u \sqrt{2-u} \, du$$

1st Approach

$$\text{let } k = 2 - u$$

$$\text{when } u=1, k=1$$

$$u=0, k=2$$

$$\frac{dk}{du} = -1$$

$$\therefore I = 2 \int_2^1 (2-k) \sqrt{k} (-dk) \quad \text{--- (1)}$$

$$= 2 \int_1^2 2k^{\frac{1}{2}} - k^{\frac{3}{2}} \, dk$$

$$= 2 \left[\frac{2k^{\frac{3}{2}}}{\frac{3}{2}} - \frac{k^{\frac{5}{2}}}{\frac{5}{2}} \right]_1^2 \quad \text{--- (1)}$$

$$= 2 \left[\frac{4k^{\frac{3}{2}}}{3} - \frac{2k^{\frac{5}{2}}}{5} \right]_1^2$$

$$= 2 \left[\left(\frac{8\sqrt{2}}{3} - \frac{8\sqrt{2}}{5} \right) - \left(\frac{4}{3} - \frac{2}{5} \right) \right]$$

$$= 2 \left[\frac{40\sqrt{2} - 24\sqrt{2}}{15} - \left(\frac{20-6}{15} \right) \right]$$

$$= 2 \left(\frac{16\sqrt{2}}{15} - \frac{14}{15} \right)$$

$$= \frac{32\sqrt{2} - 28}{15} \quad \text{--- (1)}$$

2nd Approach:

$$\text{Let } u = 2\sin^2\theta \quad \text{when } u=1, \theta=\frac{\pi}{4} \\ u=0, \theta=0$$

$$\frac{du}{d\theta} = 4\sin\theta\cos\theta$$

$$\therefore I = 2 \int_0^{\frac{\pi}{4}} \sqrt{2-2\sin^2\theta} \cdot 2\sin^2\theta \cdot 4\sin\theta\cos\theta d\theta$$

$$= 16\sqrt{2} \int_0^{\frac{\pi}{4}} \sin^3\theta \cos\theta d\theta \quad \text{--- (1)}$$



The rest follows from the original solution when we made the substitution $x = 4\sin^4\theta$ initially.

3rd Approach:

$$I = 2 \int_0^1 u\sqrt{2-u} du \quad \text{Integration by parts}$$

$$= 2 \int_0^1 \frac{d}{du} \left(-\frac{(2-u)^{3/2}}{3/2} \right) u du$$

$$= \frac{4}{3} \left[-u(2-u)^{3/2} \right]_0^1 + \frac{4}{3} \int_0^1 (2-u)^{3/2} du \quad \text{--- (1)}$$

$$= -\frac{4}{3} + \frac{4}{3} \left[-\frac{(2-u)^{5/2}}{5/2} \right]_0^1 \quad \text{--- (1)}$$

$$= -\frac{4}{3} - \frac{8}{15} \left[(2-u)^{5/2} \right]_0^1$$

$$= -\frac{4}{3} - \frac{8}{15} (1 - 4\sqrt{2})$$

$$= -\frac{20}{15} - \frac{8}{15} + \frac{32\sqrt{2}}{15}$$

$$= \frac{32\sqrt{2} - 28}{15} \quad \text{--- (1)}$$

4th Approach:

$$I = 2 \int_0^1 u \sqrt{2-u} + 2 \sqrt{2-u} - 2 \sqrt{2-u} \, du$$

$$= 2 \int_0^1 (u-2) \sqrt{2-u} + 2 \sqrt{2-u} \, du \quad \text{--- (1)}$$

$$= 2 \int_0^1 2(2-u)^{\frac{1}{2}} - (2-u)^{\frac{3}{2}} \, du$$

$$= 2 \left[\frac{2(2-u)^{\frac{3}{2}}}{\frac{3}{2}} + \frac{(2-u)^{\frac{5}{2}}}{\frac{5}{2}} \right]_0^1 \quad \text{--- (1)}$$

$$= 4 \left[\left(\frac{1}{5} - \frac{2}{3} \right) - \left(\frac{4\sqrt{2}}{5} - \frac{4\sqrt{2}}{3} \right) \right]$$

$$= 4 \left[\frac{-7}{15} + \frac{8\sqrt{2}}{15} \right]$$

$$= \frac{32\sqrt{2}-28}{15} \quad \text{--- (1)}$$

2016 Extension 2 Term 1 Task Q 8

Part (a)

(i)

Note that $\forall z \in \mathbb{C}, |z| \geq 0$

$$|z - 2| \geq 0 \Rightarrow \operatorname{Re}\left(z - \frac{1}{2}\right) \geq 0$$

$$z = x + iy \Rightarrow \operatorname{Re}\left(x + iy - \frac{1}{2}\right) \geq 0$$

$$x - \frac{1}{2} \geq 0$$

$$x \geq \frac{1}{2} \quad \text{(1 mark)}$$

Comments:

Students needed to demonstrate they knew $|z - 2| \geq 0$ and that, therefore, the real component was also greater than zero.

(ii)

$$|x - 2 + iy| = 2\left(x - \frac{1}{2}\right)$$

$$(x - 2)^2 + y^2 = 4\left(x^2 - x + \frac{1}{4}\right) \quad \text{(1 mark)}$$

$$x^2 - 4x + 4 + y^2 = 4x^2 - 4x + 1$$

$$3x^2 - y^2 = 3 \quad \text{(1 mark)}$$

(iii)

Locus is an hyperbola with vertex at (1,0), domain restricted (noting above result) to $x \geq \frac{1}{2}$. It's therefore the right-hand branch only. **(1 mark)**

Comments:

The mark was not awarded unless the domain restriction was clearly stated.

(iv)

$$\text{Asymptotes: } y = \pm\sqrt{3}x$$

(1 mark)

Comments:

The asymptotes had to be clearly shown and labelled for the mark to be awarded. A mark for carry-forward error was awarded if the student had failed to show the domain restriction in part (iii), and had proceeded to otherwise correctly graph the entire hyperbola.

(v)

To find the boundaries of the argument range, find the angles of inclination of the asymptotes:

$$\tan \theta = \pm \sqrt{3}$$

$$\theta = \pm \frac{\pi}{3}$$

$$-\frac{\pi}{3} < \arg z < \frac{\pi}{3} \quad (1 \text{ mark})$$

Comments:

The student needed to write the argument as a strict inequality.

If the student had carried forward the error of not restricting the domain in part (iii), then a mark was awarded for finding the domain for the left-hand arm of the hyperbola: $\frac{2\pi}{3} < \arg z < \frac{4\pi}{3}$

Part (b)

(i)

Method 1:

$$\text{Let } m = x^2, \quad x = \sqrt{m}$$

$$3x^3 - 9x^2 + 6x + 2 = 0$$

$$3m\sqrt{m} - 9m + 6\sqrt{m} + 2 = 0 \quad (1 \text{ mark})$$

$$\sqrt{m}(3m + 6) = 9m - 2$$

$$m(9m^2 + 36m + 36) = 81m^2 - 36m + 4 \quad (1 \text{ mark})$$

$$9m^3 - 45m^2 + 72m - 4 = 0$$

(It was unnecessary to change the variable back)

Method 2:

Find each of the following:

$$\alpha^2 + \beta^2 + \gamma^2 = 5$$

$$\alpha^2\beta^2\gamma^2 = \frac{4}{9}$$

$$\alpha^2\beta^2 + \beta^2\gamma^2 + \alpha^2\gamma^2 = 8 \quad (1 \text{ mark})$$

$$x^3 - 5x^2 + 8x - \frac{4}{9} = 0 \quad (1 \text{ mark})$$

(ii)

$$\alpha^3\beta\gamma + \alpha\beta^3\gamma + \alpha\beta\gamma^3 = \alpha\beta\gamma(\alpha^2 + \beta^2 + \gamma^2) \quad (1 \text{ mark})$$

$$\alpha\beta\gamma = -\frac{2}{3}$$

$$\alpha^2 + \beta^2 + \gamma^2 = 5$$

$$\therefore \alpha^3\beta\gamma + \alpha\beta^3\gamma + \alpha\beta\gamma^3 = -\frac{2}{3} \times 5$$

$$= -\frac{10}{3} \text{ (1 mark)}$$

Comments:

If Method 2 was used in part (ii) and an error was made in finding the sum or product of the roots, and that error was brought forward here, then a mark for carry forward error was awarded.

Part (c)

(i)

$$\text{Let } x = \frac{2}{3} \cos \theta$$

$$27 \left(\frac{2}{3} \cos \theta \right)^3 - 9 \left(\frac{2}{3} \cos \theta \right) + 1$$

$$= 27 \left(\frac{8}{27} \cos^3 \theta \right) - 6 \cos \theta + 1 \text{ (1 mark)}$$

$$= 8 \cos^3 \theta - 6 \cos \theta + 1$$

$$= 2 \cos 3\theta + 1, \text{ since } \cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$$

$$\therefore 2 \cos 3\theta + 1 = 0$$

$$\cos 3\theta = -\frac{1}{2} \text{ (1 mark)}$$

$$3\theta = \pm \frac{2\pi}{3} + 2k\pi, \quad k = 0, 1, 2$$

$$\theta = \pm \frac{2\pi}{9} + \frac{2k\pi}{3}$$

$$\theta = \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}$$

$$x = \frac{2}{3} \cos \frac{2\pi}{9}, \frac{2}{3} \cos \frac{4\pi}{9}, \frac{2}{3} \cos \frac{8\pi}{9} \text{ (or equivalent values, noting that } \cos \theta \text{ is an even function) (1 mark)}$$

Comments:

Many students made computational errors when finding the angle.

(ii)

$$\left(\cos \frac{\pi}{9} \right) \left(\cos \frac{3\pi}{9} \right) \left(\cos \frac{5\pi}{9} \right) \left(\cos \frac{7\pi}{9} \right) = \left(-\cos \frac{8\pi}{9} \right) \left(\frac{1}{2} \right) \left(-\cos \frac{4\pi}{9} \right) \left(-\cos \frac{2\pi}{9} \right) \text{ (1 mark)}$$

$$= -\frac{1}{2} \left(\cos \frac{2\pi}{9} \right) \left(\cos \frac{4\pi}{9} \right) \left(\cos \frac{8\pi}{9} \right)$$

From the equation $27x^3 - 9x + 1 = 0$,

$$\text{Product of the roots} = -\frac{1}{27}$$

$$\therefore \frac{2}{3} \cos \frac{2\pi}{9} \times \frac{2}{3} \cos \frac{4\pi}{9} \times \frac{2}{3} \cos \frac{8\pi}{9} = -\frac{1}{27} \text{ (1 mark)}$$

$$\left(\cos \frac{2\pi}{9} \right) \left(\cos \frac{4\pi}{9} \right) \left(\cos \frac{8\pi}{9} \right) = -\frac{1}{8}$$

$$\therefore \left(\cos \frac{\pi}{9} \right) \left(\cos \frac{3\pi}{9} \right) \left(\cos \frac{5\pi}{9} \right) \left(\cos \frac{7\pi}{9} \right) = -\frac{1}{2} \times -\frac{1}{8}$$

$$= \frac{1}{16} \text{ (1 mark)}$$

Comments:

Students were required to show the equivalence of the product of the roots derived directly from the equation, and the roots found in part (i).

Note that if the roots were determined incorrectly in part (i) and the logic of part (ii) was **obviously** otherwise sound, then a mark was awarded for carry-forward error.

(i) GRADIENT at $P(3\cos\theta, 2\sin\theta)$:

$$\frac{dy}{dx} = \frac{-4x}{9y} = \frac{-4}{9} \cdot \frac{3\cos\theta}{2\sin\theta} = -\frac{2\cos\theta}{3\sin\theta}$$

EQ of tangent at P :

$$y - 2\sin\theta = -\frac{2\cos\theta}{3\sin\theta} (x - 3\cos\theta)$$

$$3\sin\theta y - 6\sin^2\theta = 6\cos^2\theta - 2\cos\theta x$$

$$3\sin\theta y + 2\cos\theta x = 6(\cos^2\theta + \sin^2\theta) = 6$$

$$\frac{x}{3}\cos\theta + \frac{y}{2}\sin\theta = 1$$

$$\star \text{ or } 2\cos\theta x + 3\sin\theta y - 6 = 0$$

(ii) EQ of OQ : $y = -\frac{2\cos\theta}{3\sin\theta} x$ meets ellipse

$$\frac{x^2}{9} + \frac{4\cos^2\theta}{4 \cdot 9 \sin^2\theta} x^2 = 1$$

$$x^2(1 + \cot^2\theta) = 9$$

$$x^2 = 9 \sin^2\theta$$

$$\text{But } x < 0 \quad \therefore x = -3\sin\theta, y = \frac{-2\cos\theta}{3\sin\theta} \cdot (-3\sin\theta)$$

$$\therefore Q = [-3\sin\theta, 2\cos\theta]$$

(iii) Perpendicular distance of origin to tangent at P = $\frac{|(2\cos\theta) \cdot 0 + (3\sin\theta) \cdot 0 - 6|}{\sqrt{4\cos^2\theta + 9\sin^2\theta}}$
(from eq $\star$)

$$= \frac{|-6|}{\sqrt{4\cos^2\theta + 9\sin^2\theta}}$$

$$= \frac{6}{\sqrt{(0 - 3\sin\theta)^2 + (0 - 2\cos\theta)^2}}$$

$$= \frac{6}{OQ} \quad \#$$

slope must be in parametric form

EQ of tangent at P

m for EQ of OQ
(not for slope of OQ)must show $x < 0$
correct coordinates for Qmust show clearly $\perp$ dist
from which pt
to which line

simplifying expression

must show

$$OQ = \sqrt{(0 - 3\sin\theta)^2 + (0 - 2\cos\theta)^2}$$

alternatively can use
from Q $(-3\sin\theta, 2\cos\theta)$
to tangent

Alternatively:

$$\text{Eq } OQ: y = -\frac{2\cos\theta}{3\sin\theta}x$$

$$\therefore 2\cos\theta x + 3\sin\theta y = 0$$

Perp. distance from P to OQ:

$$= \frac{|2\cos\theta(3\cos\theta) + 3\sin\theta(2\sin\theta)|}{\sqrt{(2\cos\theta)^2 + (3\sin\theta)^2}}$$

$$= \frac{|6(\cos^2\theta + \sin^2\theta)|}{\sqrt{4\cos^2\theta + 9\sin^2\theta}}$$

$$= \frac{6}{\sqrt{(0-2\cos\theta)^2 + (0-3\sin\theta)^2}}$$

$$= \frac{6}{OQ}$$

$$\text{iv) Area } \triangle OPQ = \frac{1}{2} \cdot \frac{6}{OQ} \cdot OQ = 3 \text{ square unit}$$

$\therefore$ independent of position of P

$$\text{v) } PQ = \sqrt{(3\cos\theta + 3\sin\theta)^2 + (2\sin\theta - 2\cos\theta)^2}$$

$$= \sqrt{9\cos^2\theta + 18\cos\theta\sin\theta + 9\sin^2\theta + 4\cos^2\theta - 8\cos\theta\sin\theta + 4\sin^2\theta}$$

$$= \sqrt{13 + 10\cos\theta\sin\theta}$$

$$PQ = 13 + 5\sin 2\theta$$

$$\text{max } PQ = 18$$

$$\text{min } PQ = 8$$

$$\text{when } \sin 2\theta = 1$$

$$\text{when } \sin 2\theta = -1$$

$$\text{vi) slope of tangent at Q} = \frac{-\frac{4}{9}x}{y} = \frac{-\frac{4}{9} \cdot \frac{-3\sin\theta}{2\cos\theta}}{2\cos\theta}$$

$$= \frac{2\sin\theta}{3\cos\theta} \therefore \text{parallel to } OP$$

$\therefore OPQT$ is a para (2 pairs of opposite sides parallel)

$$\therefore T = [3(\cos\theta - \sin\theta), 2(\sin\theta + \cos\theta)]$$

P2

Using this method
★ must show eq. of line OQ clearly & P(cosθ, 2sinθ) (using formula)

many students fudging

since answer was given.

many students did not show the details how to find OQ & lost this mark

many did not mention independent of P & some forgot $\frac{1}{2}$ for area Δ .

Some students use calculus to find max/min

both max/min must be correct to get this mark

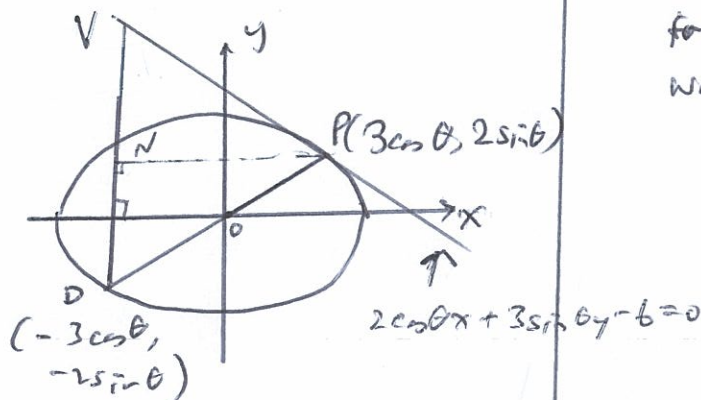
Most students try to find eq of tangent at Q & solve simultaneously with tangent at P but failed

both x, y must be correct

$$\begin{aligned}
 & 4x^2 + 9y^2 \\
 &= 4[3(\cos\theta - \sin\theta)]^2 + 9[2(\sin\theta + \cos\theta)]^2 \\
 &= 36[\cancel{\cos^2\theta} + \cancel{\sin^2\theta} - 2\cancel{\sin\theta\cos\theta} + \sin^2\theta + \\
 &= 2\cancel{\sin^2\theta} + \cancel{\cos^2\theta}] \\
 &= 36 \times 2 = 72 \quad (\sin^2\theta + \cos^2\theta = 1)
 \end{aligned}$$

v) Area of Ellipse $= 2 \times 2 \times 3\pi = 6\pi$

$$D = [-3\cos\theta, -2\sin\theta]$$



To find the coordinates of V = ?

V lies on tangent at P

when $x = -3\cos\theta$, $2\cos\theta(-3\cos\theta) + 3\sin\theta y - 6 = 0$

$$y = \frac{6 + 6\cos^2\theta}{3\sin\theta} = \frac{2 + 2\cos^2\theta}{\sin\theta}$$

$$V = \left[-3\cos\theta, \frac{2 + 2\cos^2\theta}{\sin\theta}\right]$$

$$\therefore VD = \left| \frac{2 + 2\cos^2\theta}{\sin\theta} + 2\sin\theta \right| = \left| \frac{2 + 2(\cos^2\theta + \sin^2\theta)}{\sin\theta} \right|$$

$$VD = \left| \frac{4}{\sin\theta} \right|$$

Construct $PN \perp VD$

$$PN = 6|\cos\theta|$$

$$\therefore \text{Area } \triangle PVD = \frac{1}{2} \times 6|\cos\theta| \times \left| \frac{4}{\sin\theta} \right| = \frac{12}{|\tan\theta|}$$

$$\therefore \text{Area } \triangle PVD : \text{Area Ellipse} =$$

$$= \frac{12}{|\tan\theta|} : 6\pi = 2 : \pi |\tan\theta|$$

Will Done

Students get 1 m for area of ellipse. Will done

Students get 1 m for the coordinates of $V = (-3\cos\theta, \frac{2 + 2\cos^2\theta}{\sin\theta})$ or $VD = \frac{2 + 2\cos^2\theta}{\sin\theta} + 2\sin\theta$

MATHEMATICS Extension 2: Question 10...

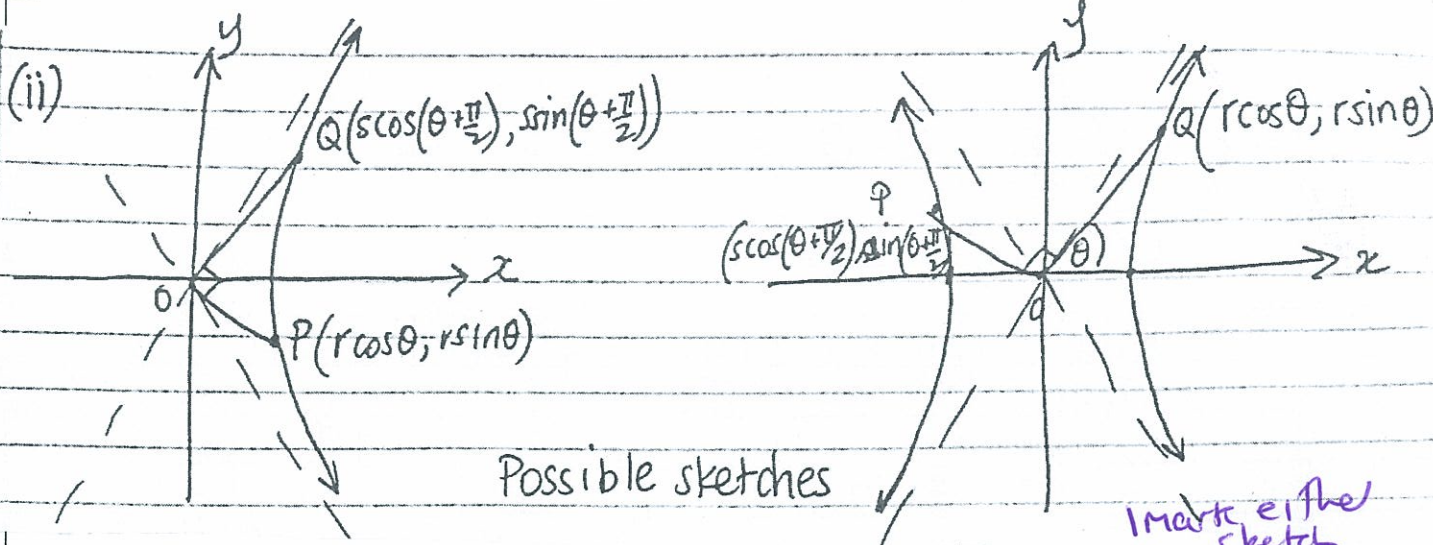
Suggested Solutions	Marks	Marker's Comments
(a) Sub $(r \cos \theta, r \sin \theta)$ and $(s \cos(\theta + \frac{\pi}{2}), s \sin(\theta + \frac{\pi}{2}))$ into curve $\frac{r^2 \cos^2 \theta}{a^2} - \frac{r^2 \sin^2 \theta}{b^2} = 1$ i.e. $\frac{\cos^2 \theta}{a^2} - \frac{\sin^2 \theta}{b^2} = \frac{1}{r^2} \quad \text{--- (1)}$ Now $s \cos(\theta + \frac{\pi}{2}) = -s \sin \theta$ $s \sin(\theta + \frac{\pi}{2}) = s \cos \theta$ $\frac{s^2 \sin^2 \theta}{a^2} - \frac{s^2 \cos^2 \theta}{b^2} = 1$ $\frac{\sin^2 \theta}{a^2} - \frac{\cos^2 \theta}{b^2} = \frac{1}{s^2} \quad \text{--- (2)}$	1	Had to change $\cos(\theta + \frac{\pi}{2})$ to $-\sin \theta$ to gain first mark.
(ii) $\frac{1}{r^2} + \frac{1}{s^2} = \frac{\sin^2 \theta + \cos^2 \theta}{a^2} - \frac{\sin^2 \theta + \cos^2 \theta}{b^2}$ $= \frac{1}{a^2} - \frac{1}{b^2}$	1	Too many students didn't explain how they got last step
$P(r \cos \theta, r \sin \theta)$ and $Q(s \cos(\theta + \frac{\pi}{2}), s \sin(\theta + \frac{\pi}{2}))$ $\frac{1}{OP^2} + \frac{1}{OQ^2} = \frac{1}{(r \cos \theta)^2 + (r \sin \theta)^2} + \frac{1}{(s \sin \theta)^2 + (s \cos \theta)^2}$ $= \frac{1}{r^2} + \frac{1}{s^2}$ $= \frac{1}{a^2} - \frac{1}{b^2} \quad \text{from (i)}$ which is independent of the position of r and θ.	1	Student needed to clearly explain what OP and OQ were not just state $= r$ and s .

Ex+ 2 MATHEMATICS: Question 10

Suggested Solutions

Marks

Marker's Comments



$$(b) x^2y + xy^2 + 16 = 0$$

$$x^2 \frac{dy}{dx} + 2xy + 2xy \frac{dy}{dx} + y^2 = 0$$

$$\frac{dy}{dx} = -\frac{y(y+2x)}{x(x+2y)}$$

$$M_T = -1$$

$$\therefore -1 = -\frac{y(y+2x)}{x(x+2y)}$$

$$x^2 + 2xy = y^2 + 2xy$$

$$x^2 = y^2$$

$$x = \pm y$$

$$\text{if } x = y$$

$$x^3 + x^3 + 16 = 0$$

$$2x^3 = -16$$

$$x^3 = -8$$

$$y = -2$$

$$y = -x$$

$$-x^3 + x^3 + 16 = 0$$

no solution

2

1 for 2 correct terms
2 for correct expression

1

poor algebra
many students
lost negative sign.

1

needed to look
at both
cases

EXT 2 MATHEMATICS: Question 10...

Suggested Solutions

Marks

Marker's Comments

$$c) (i) \int \ln(1+x) \frac{d}{dx} (1+x) dx$$

$$= (1+x) \ln(1+x) - \int 1+x \times \frac{1}{1+x} dx$$

$$= \underline{(1+x) \ln(1+x) - x + C}$$

$$(ii) \int_0^1 x^n \frac{d}{dx} [(1+x) \ln(1+x) - x] dx$$

$$= \left[x^n (1+x) \ln(1+x) - x \right]_0^1 - n \int_0^1 x^{n-1} [(1+x) \ln(1+x) - x] dx$$

$$= 2 \ln 2 - 1 - n \int_0^1 x^{n-1} \ln(1+x) + x^n \ln(1+x) - x^n dx$$

$$= 2 \ln 2 - 1 - n I_{n-1} - n I_n + n \int_0^1 x^n dx$$

$$= 2 \ln 2 - 1 - n I_{n-1} - n I_n + \left[\frac{n x^{n+1}}{n+1} \right]_0^1$$

$$\therefore I_n = 2 \ln 2 - 1 - n I_{n-1} - n I_n + \frac{n}{n+1}$$

$$(n+1) I_n = 2 \ln 2 - \frac{1}{n+1} - n I_{n-1}$$

Number of different methods

1

1

1 for correct
✓ those who
close x^n made
little progress.
why is part (i)
there?

1

needed to see
 $-1 + \frac{n}{n+1}$ to get
2nd mark.
many students
made 2 errors

Ext 2 MATHEMATICS: Question 10

Suggested Solutions

Marks

Marker's Comments

(iii) When $n=0$

$$I_0 = 2 \ln 2 - 1$$

~~2 I_1~~

$$2I_1 = 2 \ln 2 - \frac{1}{2} - I_0$$

$$3I_2 = 2 \ln 2 - \frac{1}{3} - 2I_1$$

$$= 2 \ln 2 - \frac{1}{3} - \left[2 \ln 2 - \frac{1}{2} - 2(2 \ln 2 - 1) \right]$$

$$= 2 \ln 2 - \frac{1}{3} + \frac{1}{2} - 4 \ln 2 + 4$$

$$= 2 \ln 2 - \frac{5}{6}$$

$$4I_3 = 2 \ln 2 - \frac{1}{4} - 3I_2$$

$$= 2 \ln 2 - \frac{1}{4} - \left[2 \ln 2 - \frac{5}{6} \right]$$

$$= -\frac{1}{4} + \frac{5}{6}$$

$$= \frac{7}{12}$$

1

1 mark for I_0 and some progress to calculate $3I_2$

• Many students made careless errors with negative sign.

1

2nd mark for correct $4I_3$.

3x2 MATHEMATICS: Question 1.0

Suggested Solutions

Marks

Marker's Comments

$$(n+1)I_n = \left(2\ln 2 - \frac{1}{n+1}\right) - \left(2\ln 2 - \frac{1}{1}\right)$$

$$+ \dots (-1)^{n+1} \left(2\ln 2 - \frac{1}{2}\right) + (-1)^n \left(2\ln 2 - 1\right)$$

If n is odd even number of terms so $2\ln 2 - 2\ln 2 + 2\ln 2 - 2\ln 2 \dots = 0$

$$\therefore (n+1)I_n = -\frac{1}{n+1} + \frac{1}{1} - \dots - \frac{1}{2} + 1$$

If n is even odd number of terms $2\ln 2 - 2\ln 2 + 2\ln 2 - 2\ln 2 \dots = 2\ln 2$

$$(n+1)I_n = 2\ln 2 - \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{n+1}\right)$$

1

1 mark for building up pattern.

= 0

Students needed clearly explain why $2\ln 2$ when ~~even~~ and no $2\ln 2$ when odd.

1

very students did this. Some proved by induction which was it needed but needed to prove for both odd and even